

Functional Annotation of *glcD* (PP_3745, UniProt Q88GH8) in *Pseudomonas putida* KT2440

1. Summary / Answer to the Research Question

glcD (PP_3745; UniProt Q88GH8) encodes the FAD-linked catalytic subunit of the bacterial glycolate oxidase / glycolate dehydrogenase (EC 1.1.99.14; KEGG Orthology K00104). Its primary function is to catalyze the flavin-dependent oxidation of **glycolate (2-hydroxyacetate) to glyoxylate**. GlcD does not act alone: it is one of three subunits of a heteromeric enzyme, GlcDEF, in which GlcD and GlcE are FAD-binding flavoprotein subunits and GlcF is an iron–sulfur subunit that relays electrons to the membrane quinone pool. The enzyme is a **membrane-associated respiratory dehydrogenase** (electrons pass to quinones/ respiratory chain, not directly to O₂ — hence the "1.1.99, unknown/other acceptor" classification). In *P. putida* KT2440 this reaction is the committed step of **glycolate catabolism**, feeding glyoxylate into central carbon metabolism (glyoxylate carboligase and glycerate/ glyoxylate-cycle routes) and detoxifying glycolate generated during growth on C2 compounds such as ethylene glycol.

Gene identity was verified and is unambiguous: the gene symbol *glcD*, the protein description (glycolate oxidase FAD-linked subunit, EC 1.1.99.14), the organism (*P. putida* KT2440), and the FAD-bd_PCMH domain set given in the UniProt record all match the coherent body of literature and database evidence assembled below.

2. Gene / Protein Identity and Verification

Attribute	Value	Source
Gene symbol	<i>glcD</i>	UniProt Q88GH8; KEGG ppu:PP_3745
Locus tag	PP_3745	KEGG; genome AE015451 (AAN69340.1)
Protein length	499 aa (53.5 kDa)	UniProt Q88GH8
Enzyme	Glycolate dehydrogenase, FAD-linked subunit; EC 1.1.99.14	KEGG KO K00104
Cofactor	FAD	UniProt (ChEBI:57692)
Pathway	ppu00630 Glyoxylate and dicarboxylate metabolism	KEGG
Domains	FAD_binding_4 (PF01565) + FAD-oxidase_C (PF02913); FAD-bd_PCMH; TIGR00387 (glcD); COG0277	UniProt/InterPro
Orthology	80.8% identity to <i>E. coli</i> GlcD (P0AEP9)	This work (global alignment)

The gene symbol, the enzyme description, the FAD-binding PCMH domain architecture specified in the target record, and the KEGG orthology assignment (K00104, EC 1.1.99.14) are fully mutually consistent. There is **no ambiguity** in this identification.

3. Primary Function: Reaction Catalyzed and Substrate Specificity

3.1 Reaction

GlcD is the flavin-containing catalytic subunit that carries out:



i.e., oxidation of the C2 α -hydroxyl of glycolate (the smallest 2-hydroxyacid) to the carbonyl of glyoxylate, with hydride transfer to enzyme-bound FAD and subsequent re-oxidation of the flavin by downstream electron carriers.

3.2 Enzyme is a three-subunit complex (GlcDEF)

Classic genetic dissection of the *E. coli* *glc* locus showed that GlcD and GlcE resemble flavoenzymes while GlcF resembles iron–sulfur proteins, and that a cassette insertion in *glcD*, *glcE*, or *glcF* each abolishes glycolate oxidase activity — establishing that all three are required subunits of the enzyme (Pellicer et al. 1996,).

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The *P. putida* GlcD therefore functions as the FAD-linked catalytic partner within a GlcDEF heterotrimer, with GlcF providing the [Fe–S] electron-relay.

3.3 Substrate specificity

GlcD belongs to the **FAD-dependent 2-hydroxyacid oxidase/dehydrogenase (VAO/PCMH) superfamily**, which includes glycolate oxidase (GOX), L-lactate oxidase, and D-lactate dehydrogenase. Within this superfamily, glycolate-vs-lactate specificity is governed by a small number of active-site residues: Hackenberg et al. (2011,)

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showed cyanobacterial/algal GlcD-type enzymes catalyze glycolate oxidation, and that converting the active site of a related lactate oxidase (No-LOX) to that of Arabidopsis GOX2 reversed its lactate/glycolate activity ratio. The bacterial GlcDEF enzyme is annotated and physiologically dedicated to **glycolate** as its preferred substrate (EC 1.1.99.14).

3.4 Evolutionary support (functional transfer)

A gap-free global alignment shows **80.8% amino-acid identity over all 499 residues** between *P. putida* GlcD (Q88GH8) and the experimentally characterized *E. coli* GlcD (P0AEP9). This is strict 1:1 orthology (not merely shared family membership), which permits high-confidence transfer of the *E. coli* experimental annotation and upgrades the UniProt "putative/predicted" (evidence level 4) label to a well-supported functional assignment.

4. Subcellular Localization and Site of Action

EC 1.1.99.14 is a **dehydrogenase** (sub-subclass 1.1.99 = "acting on CH–OH with unknown/other acceptors"), not an O₂-dependent oxidase. Biochemical study of a glycolate-oxidizing bacterium demonstrated that glycolate dehydrogenase activity reduces artificial electron acceptors (benzyl viologen, methylene blue, DCPIP, ferricyanide, water-soluble quinones) and is recovered as a **membrane-bound** activity with **menaquinones as electron carriers** (Friedrich & Schink 1993,

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Consistent with this, InterPro/GO assign GlcD to the "**glycolate oxidase complex**" (GO:0009339) with FAD binding.

Interpretation of localization: GlcD carries out its reaction as part of a membrane-associated GlcDEF complex at the cytoplasmic face of the inner membrane. Electrons flow substrate → GlcD-FAD → GlcE-FAD → GlcF [Fe-S] → membrane quinone (ubiquinone/menaquinone) pool, coupling glycolate oxidation to the aerobic respiratory chain. This mechanism explains the "dehydrogenase" (quinone acceptor) rather than "oxidase" (O₂ acceptor) designation.

4.1 Structure-based evidence for a catalytically competent flavoenzyme

Motif analysis of the 499-aa GlcD sequence identifies the conserved hallmarks of the FAD-linked oxidase (VAO/PCMH; D-lactate-dehydrogenase-like) superfamily, all present at identical positions in the experimentally validated *E. coli* GlcD ortholog:

Motif	<i>P. putida</i> position	Role
GAGTG	86–90	N-terminal FAD-anchoring glycine-rich loop
GGNVAEN	~157	FAD_binding_4 (PF01565) domain signature
HAGDGNMH	386–393	C-terminal FAD-oxidase domain motif
E-H431-G-V-G-R-E (EHGVGRE)	430–436	Catalytic motif with invariant His431

The catalytic motif is essentially identical between the two proteins (*P. putida* SITGEHGVGREKINQ vs *E. coli* SISGEHGIGREKINQ; **His431 invariant**). In this flavoenzyme family the conserved active-site histidine acts as the catalytic base that deprotonates the substrate C2 α-hydroxyl during hydride transfer to FAD. Conservation of both the FAD-binding loop and the

catalytic His between *P. putida* GlcD and the genetically validated *E. coli* enzyme is structure/sequence-based evidence that GlcD is a **catalytically competent** FAD-dependent glycolate dehydrogenase subunit, not a degenerate/pseudo-enzyme.

5. Genomic Context, Regulation, and Pathway

5.1 Operon

PP_3745 lies within a conserved **glcC–glcDEFG** cluster in *P. putida* KT2440 (KEGG):

Locus	Gene	KO	Product
PP_3744	<i>glcC</i>	K11474	GntR-family transcriptional activator of the <i>glc</i> operon (divergently oriented)
PP_3745	<i>glcD</i>	K00104	Glycolate dehydrogenase, FAD-linked subunit (target)
PP_3746	<i>glcE</i>	K11472	Glycolate dehydrogenase, FAD-binding subunit
PP_3747	<i>glcF</i>	K11473	Glycolate dehydrogenase, iron–sulfur subunit
PP_3748	<i>glcG</i>	K11477	<i>glc</i> operon protein of unknown function

The genes are contiguous and co-oriented (*glcD* 4,274,044–4,275,543; *glcE* 4,275,543–4,276,595; *glcF* 4,276,605–4,277,840; *glcG* 4,277,833–4,278,231), mirroring the *E. coli glc* locus first characterized by Pellicer et al. (1996), where *glcC* is divergent from *glcDEFG* and encodes the pathway regulator. Expression of the structural genes is glycolate-inducible via GlcC.

5.2 Metabolic pathway and physiological role in *P. putida*

The product **glyoxylate** is assimilated through the glyoxylate carboligase (Gcl) pathway and glycerate/glyoxylate-cycle routes into gluconeogenesis and the TCA cycle (KEGG *ppu00630*). *P. putida* KT2440 oxidizes ethylene glycol via the toxic intermediates glycolaldehyde and glycolate; Franden et al. (2018,

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showed that overexpression of the **glcDEF glycolate oxidase operon "removes the glycolate bottleneck and minimizes the production of these toxic intermediates,"** enabling efficient

growth on this C2 substrate. Li et al. (2019,

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similarly placed glycolate oxidation within the "initial oxidations of ethylene glycol to glyoxylate" that must be balanced for C2 growth.

Thus GlcD's precise physiological role is the committed oxidation of glycolate to glyoxylate, simultaneously (i) enabling use of glycolate (and upstream ethylene glycol/glycolaldehyde) as carbon and energy sources and (ii) preventing accumulation of cytotoxic C2 intermediates.

6. Supported and Refuted Hypotheses

Supported: - H1 (identity): *glcD*/PP_3745 encodes the FAD-linked subunit of glycolate dehydrogenase (EC 1.1.99.14). ✓ Multiple concordant lines (UniProt, KEGG K00104, domains, 80.8% orthology to characterized *E. coli* GlcD). - H2 (reaction/substrate): catalyzes glycolate → glyoxylate; glycolate is the physiological substrate. ✓ - H3 (quaternary structure): acts as part of a GlcDEF three-subunit complex (FAD + FAD + [Fe-S]). ✓ (*E. coli* genetics; conserved operon). - H4 (localization/mechanism): membrane-associated, quinone-linked respiratory dehydrogenase (not O₂ oxidase). ✓ - H5 (pathway): routes glycolate/C2 carbon into central metabolism via glyoxylate; detoxifies glycolate. ✓

Refuted / disfavored: - That GlcD is a soluble, O₂-dependent glycolate *oxidase* like eukaryotic peroxisomal GOX — refuted: bacterial GlcDEF is a quinone-linked, membrane-associated dehydrogenase (EC 1.1.99.14). - That GlcD is a standalone catalytic enzyme — refuted: it requires GlcE and GlcF for activity.

7. Evidence Quality, Limitations, and Future Directions

Strengths of the annotation. The functional assignment rests on (i) direct genetic evidence in the *E. coli* ortholog (subunit knockouts abolish activity), (ii) very high, gap-free sequence orthology (80.8% over 499 aa), (iii) consistent domain/family/EC/KO assignments, and (iv) direct experimental demonstration of the *glcDEF* operon's glycolate-detoxifying role in *P. putida* KT2440 itself.

Limitations. No structure, purified-enzyme kinetics, or targeted mutant has yet been reported for the *P. putida* GlcD protein specifically; substrate specificity and membrane topology are inferred from orthologs and family biochemistry. Whether FAD is covalently or non-covalently bound in GlcD (versus GlcE) has not been experimentally resolved for this protein. The precise in vivo electron acceptor (ubiquinone vs menaquinone) in *P. putida* is inferred.

Future directions. Purification/reconstitution of the *P. putida* GlcDEF complex to measure kcat/KM for glycolate versus other 2-hydroxyacids; a clean PP_3745 deletion to confirm loss of growth on glycolate/ethylene glycol; and structural determination (or AlphaFold-guided mutagenesis) of the FAD and substrate-binding pockets to define specificity-determining residues.

8. Key References

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- Database records: UniProt Q88GH8; KEGG ppu:PP_3745 (K00104, EC 1.1.99.14); InterPro FAD-bd_PCMH; NCBIfam TIGR00387 (glcD); COG0277.

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